MILITARY SPECIFICATION

INSIGNIA, RANK, OFFICERS, SHOULDER MARK, NAVY (EMBROIDERED)

This specification is approved for use by the Navy Clothing and Textile Research Facility, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

.1. Scope. This specification covers the requirements for soft shoulder marks with embroidered insignia.

1.2 Classification. The shoulder marks shall be one type and size only in the following classes (see 6.2):

Class 1 - Officers of Flag Rank
Class 2 - Commissioned Officers below Flag Rank
Class 3 - Commissioned Warrant and Warrant Officers

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8455

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SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton

MILITARY

MIL-L-17507	- Lace, Ornamental
MIL-E-17568	- Embroidery Materials, Metallic and Synthetic Metallic
MIL-C-21115	- Cloth, Tropical, Wool, Polyester/Wool
MIL-T-43548	- Thread, Polyester, Cotton-Covered

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Association ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Color Association of the United States, Inc.
DOD Standard Color Card of Official Standardized
Shades for Heraldic Yarns for Embroidery and Ribbons - 1974

(Copies of the standard noted above may be obtained from the Color Association of the United States, Inc., 24 East 38th Street, New York, NY 10016.)

3. REQUIREMENTS

3.1 Guide sample. Samples, when available, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample, in which case this document shall govern.

3.1.1 Approved contractor sample. When a guide sample is not available all requirements specified in this document and as exhibited by the approved Contractor's sample shall apply (see 6.4).

3.2 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval.

3.3 Materials

3.3.1 Cloth, polyester/wool. The material used for the outer covering of the shoulder mark shall be a 10 ounce plain weave, polyester/wool blend fabric conforming to type III, class 3 of MIL-C-21115. The cloth shall be dyed to match shade Blue 3346.

3.3.2 Cloth, fusible. The fusible material used for interlining the outer covering of shoulder mark shall be suitable for the purpose intended. The fusible cloth shall have a maximum thickness of 0.030 inches and when dyed shall have good colorfastness to wet and dry crocking when tested as specified in 4.3.1.

3.3.3 Cloth, laminated. The basic material for fabricating the shoulder marks shall be the two fabrics described in 3.3.1 and 3.3.2 fused together with heat and pressure, forming a laminated material that shall conform to the requirements specified in table I when tested as specified in 4.3.1. The laminated cloth shall show no evidence of bubbling or bleed thru.

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Table I - Laminated Fabric Requirements

Characteristic		Requirement
Stiffness, inch lbs.		
Warp	50 (min)	89 (max)
Filling	30 (min)	57 (max)
Peel strength, ounces		
Warp direction only		23 (min)

3.3.4 Thread. The thread used for attaching lace, seaming and hemming the shoulder marks shall be cotton-covered polyester conforming to ticket number 50, 2 or 3 ply of MIL-T-43548. The thread shall be dyed to approximate the shade of the face side of the laminated material or gold lace, depending on where thread is to be used. As an alternate, the thread may be cotton conforming to Type I, IC2, ticket no. 00, 3 ply of V-T-276.

3.3.5 Lace, ornamental. The gold lace for shoulder marks shall conform to MIL-L-17507 as indicated below:

a. Class 1 shoulder marks - Type II, Classes 1 thru 5, Style A, in 1/4 inch (2 vellum) and 1 inch (2 vellum) widths.

b. Class 2 shoulder marks - type II, Classes 1 thru 5, style A in 1/8 inch (1 vellum) and 3/8 inch (2 vellum) widths

c. Class 3 shoulder marks - type II, Classes 1 thru 5, Style B, sub-style B2, Application 2 and 3, Widths 1/2 and 1/4 inch.

3.3.6 Embroidery materials. The embroidery materials shall conform to type III, classes 8 and 10 of MIL-E-17568.

3.4 Design. The soft shoulder marks shall be flat and trapezoidal shaped (see figure 1).

3.4.1 Figures. Figures are furnished for information purposes only. When inconsistencies exist between the specification and the figures, the specification shall govern.

3.5 Construction

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3.5.1 Embroidery. The device for each class of shoulder mark shall be hand or machine embroidered thru the face side of the laminated cloth employing the applicable yarn specified in 3.3.6. The embroidered device shall conform to the Government furnished guide sample (see 6.3) or the approved contractor's sample (see 6.4). Sufficient laminated cloth shall be allowed in all directions so shoulder marks can be properly hemmed, joined together, and meet finished shoulder mark measurements.

3.5.2 Stitching. Hemming and joining operations shall be at locations indicated on figure 1. Each stitchline shall have a minimum of 8 stitches per inch using stitch type 301 of FED-STD-751. Seam allowances for hemming and joining operations shall be 3/8 inch wide. Edgestitching for hems shall be 1/16 to 1/8 inch from folded edge. NOTE: If, after stitching on stripes, the background cloth becomes too stiff and bulky for turning the shoulder mark inside out, hand stitching of the back seam shall be permitted. However, when hand stitching is used to close the back seam, the seam allowances shall be stitched down first.

3.5.3 Attachment of stripes (lace). All stripes shall be stitched to the laminated cloth 1/16 inch from finished edge of lace. The various width stripes shall circumvent the width direction of the shoulder mark with each end of striping being caught in the joining seam on back side of mark.

3.6 Dimensions. The overall dimensions of finished shoulder marks shall be as indicated in figure 1. Unless otherwise specified in this document dimensional tolerances shall be as follows:

Measurement	Tolerance in inches
Length of shoulder mark	± 1/8
Width of shoulder mark (each end)	± 1/8
Stripes	
Width	± 1/32
Spacing between	± 1/16
Design (Stars, anchors, corps device)	
Overall size	± 1/16
Placement	± 1/16

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3.7 Embroidered devices. The embroidered line, staff or corps devices shall be embroidered with the gold and/or silver synthetic metallic embroidery materials specified in 3.3.6. The design of the device shall be in accordance with the guide sample. Dimensions, location, and color of devices shall be as shown in the applicable figure and as indicated below:

Class 1 and 2 The star and all corps devices shall be embroidered in gold except for the following:

Dental corps - Acorns shall be embroidered in silver
Medical Corps - Acorns shall be embroidered in silver
Judge Advocate General - Millrind shall be embroidered in silver
Civil Engineer - Acorns shall be embroidered in silver.

Class 3. All class 3 corps devices shall be embroidered in gold.

3.8 Pressing. The finshed shoulder mark shall be pressed flat into the trapezoidal shape indicated in figure 1.

3.9 Workmanship. The finished shoulder mark shall be clean, well-made, and shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification (see 4.3.1.)

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4.2 First article inspection. When required, the first article submitted in accordance with 3.2, shall be inspected as specified in 4.3.3 for compliance with design, construction, workmanship, and dimensional requirements.

4.3 Inspection. Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all requirements of referenced specifications, figures, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement document. A certificate of compliance may be submitted for the fusible cloth requirements of 3.3.2 and the laminated cloth requirements of 3.3.3. The laminated cloth shall be tested for the characteristics specified in table I in accordance with the referenced test methods of FED-STD-191 in table II. All test reports shall contain the individual values used in expressing the final result. The sample unit for testing the fusible and laminated cloths shall be a 1/4 continuous and full width yard. The lot shall be unacceptable if one or more units fail to meet any requirement specified. The sample size for testing the fusible and laminated cloths shall be as follows:

Lot Size (yards)	Sample Size (units)
800 or less	2
801 thru 22,000	3
22,001 and over	5

Table II - Testing of laminated cloth

Characteristic	Requirement Paragraph	Test Method
Thickness	3.3.2	5030
Colorfastness to crocking	3.3.2	5651 1/
Appearance	3.3.3	Visual
Stiffness	3.3.3	5206 2/
Peel strength	3.3.3	5951

1/ Method B with white test cloth
2/ Average of six samples, three tested with fusible interlining side down and three with fusible interlining side up.

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4.3.2 Examination of the end item. The defects found during the examination of the end item shall be classified in accordance with 4.3.2.1 and 4.3.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.3.2.3. The lot size shall be expressed in units of pairs of shoulder marks. The sample unit shall be one finished pair (left and right, when applicable) of shoulder marks.

4.3.2.1 Visual examination. The shoulder marks shall be examined for defects in color, design, material, construction, workmanship, and the defects classified in accordance with table II.

. Table II - Classification of defects

Examine	Defect
I Fabric, laminated	1. Hole, cut, tear, misweave, slub, knot, broken end or pick - clearly visible 2. Spot or stain 3. Material not as specified 4. Color not as specified 5. Visual evidence of bubbling, color change or bleed thru
II Stripe, gold lace	1. Type, color or pattern not as specified 2. Any scuffed or abraded area 3. Spot or stain 4. Not securely attached 5. Uneven spacing between stripes
III Seams and stitching	1. Not specified seam or stitch type 2. Less than 8 stitches per inch 3. Stitching runoff 4. Broken or missing stitches 5. Loose or tight tension, affecting appearance 6. Seams or stitch lines crooked or irregular, affecting appearance 7. Thread not an approximate match to face fabric or gold lace. 8. Edgestitching on hems not 1/16 to 1/8 inch from folded edge. 9. Seam allowance for hemming and joining operations not 3/8 inch wide.
IV Embroidered design	1. Any deviation from guide sample or approved contractor sample 2. Any design missing or mispositioned, affecting appearance 3. Poorly defined, distorted or misshaped 4. Unrelated colors or yarn visible on face 5. Surface yarns loose, tight or shifted 6. Bunched, kinked or tangled yarns on face

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Table II - Classification of defects (cont'd)

Examine	Defect
V Pressing	1. Singed or scorched 2. Creases or gloss marks 3. Misshaped by pressing

4.3.2.2 Dimensional and pairing examinations. The finished shoulder marks shall be examined for defects in accordance with the following:

(a) Any measurement deviating from the minimal dimensions and tolerances specified for the finished shoulder mark shall be scored as a defect.

(b) A poor color match of synthetic metallic yarn between paired shoulder marks shall be classified as a defect.

(c) Not paired as lefts and rights shall be classified as a defect (when applicable)

4.3.2.3 Inspection levels and acceptable quality levels. The inspection levels and the acceptable quality levels, expressed in defects per 100 units for visual and dimensional examination, shall be as follows:

	Inspection Level	AQL
For visual examination (4.3.2.1)	II	6.5
For dimensional examination (4.3.2.2)	S-1	6.5
For pairing examination (4.3.2.2)	S-4	1.5

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C (see 6.2).

5.1.1 Level A. Each pair of shoulder marks shall be placed in a snug fitting style clear polyethylene film bag. All seams and closures shall be effected by heat sealing. Excess air shall be removed from each bag before the final closure. Each sealed bag containing a pair of shoulder marks shall then be packaged in a commercial grade tuck end type folding paperboard box. The outside of each box shall measure approximately 4 1/2 inches long by 2 1/2 inches wide by 1 inch in depth.

5.1.2 Level C (Commercial packaging). The shoulder marks shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and quantity per package shall be the same as that normally used by the contractor for retail distribution.

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5.2 Packing. Packing shall be level A or C, as specified (see 6.2).

5.2.1 Level A. Shoulder marks packaged as specified in 5.1 shall be packed as specified by the procuring activity.

5.2.2 Level C (Commercial packing). Shoulder marks packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. Unless otherwise specified, marking shall be as specified by the procuring activity or in accordance with the contractor's standard commercial practice.

6. NOTES

6.1 Intended use. The embroidered soft shoulder marks are intended for wear by male and female officer personnel of the Navy as insignia of ranks.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Class required (see 1.2)
c. When First Article sample is not required (see 3.2)
d. Levels of packaging and packing required (see 5.1 and 5.2).

6.3 Guide sample. For access to a guide sample address, the Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760.

6.4 Approved contractor sample. A contractor submitting a sample for Government approval shall forward the sample to the Navy Clothing and Textile Research Facility, 21 Strathmore Rd., Natick, MA 01760.

Custodian;
Navy - NU

Preparing activity -
Navy - NU

Project No. 8455-N871

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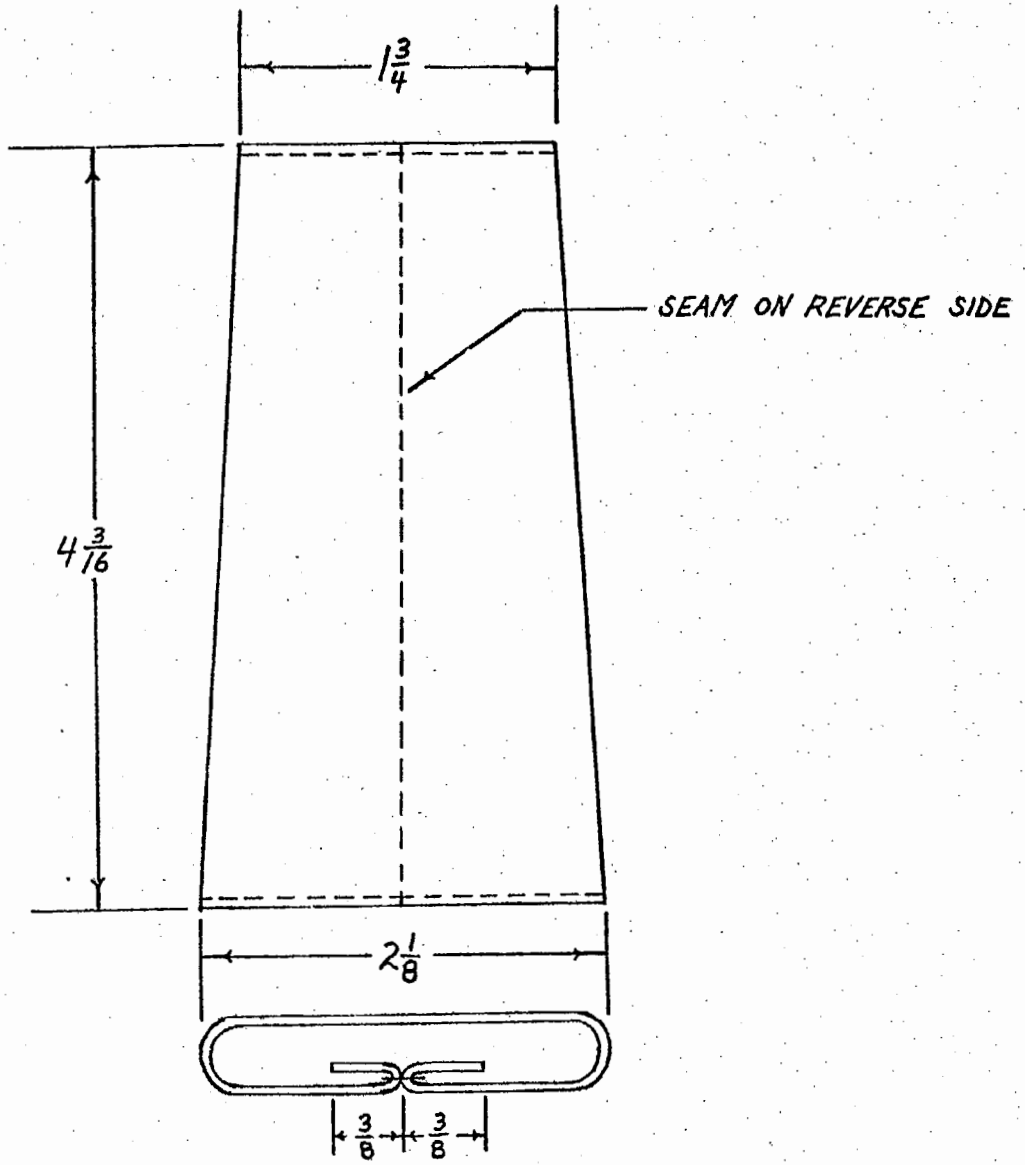
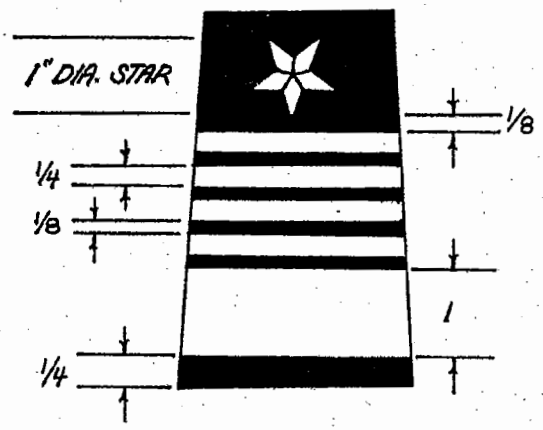
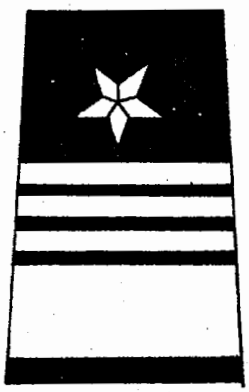


FIGURE 1 - SHOULDER MARK, SOFT

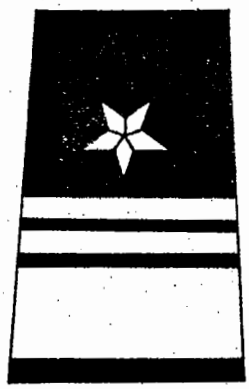
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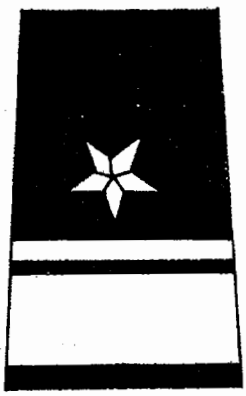
FLEET ADMIRAL



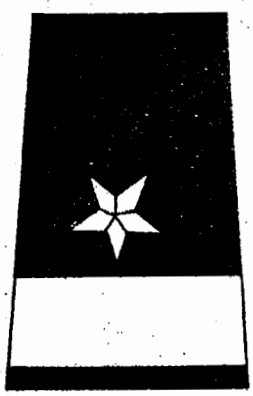
ADMIRAL



VICE ADMIRAL



REAR ADMIRAL



COMMODORE

FIGURE 2 - OFFICERS OF FLAG RANK
NOTE: WHEN APPLICABLE, CORPS DEVICE SHALL BE USED IN LIEU OF THE STAR

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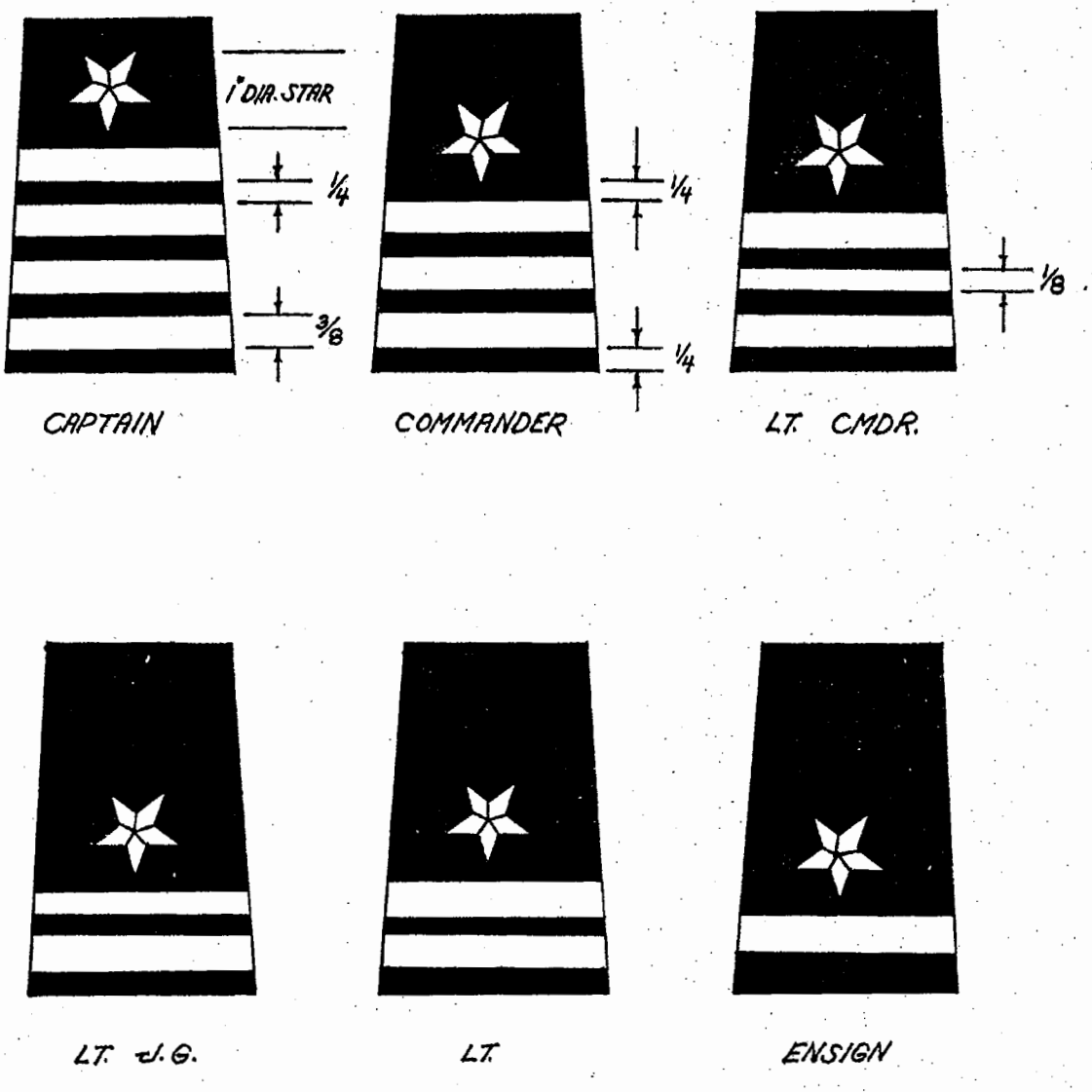
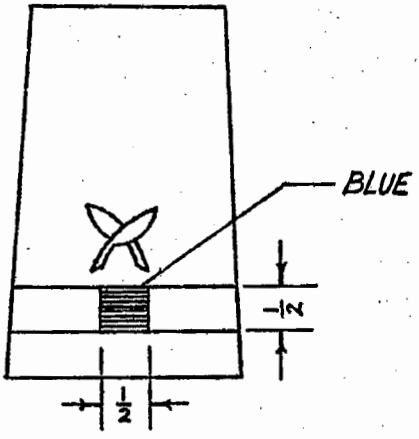
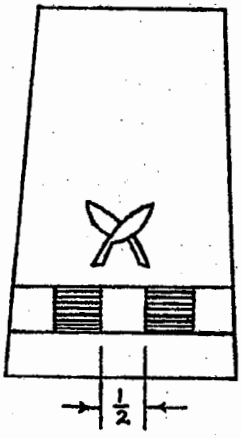


FIGURE 3 - COMMISSIONED OFFICERS BELOW FLAG RANK
NOTE: WHEN APPLICABLE, CORPS DEVICE SHALL BE USED IN LIEU OF THE STAR

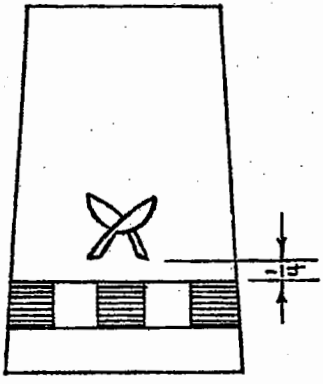
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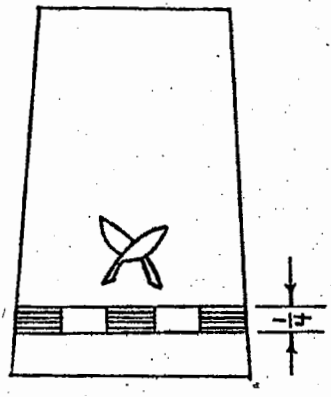
CWO (W-4)



CWO (W-3)



CWO (W-2)



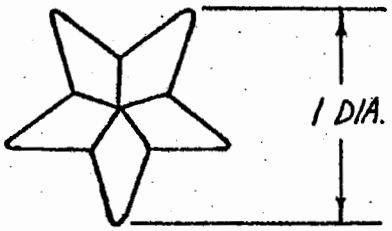
WARRANT (W-1)

CLASS 3 - COMMISSIONED WARRANT AND WARRANT OFFICERS

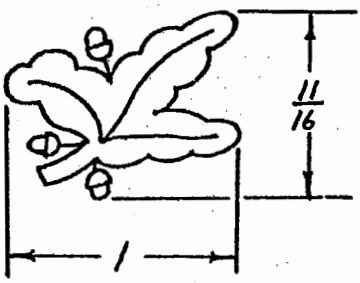
NOTE: SEE FIGURES 7, 8, AND 9 FOR CLASS 3 LINE AND STAFF CORPS DEVICES

FIGURE - 4

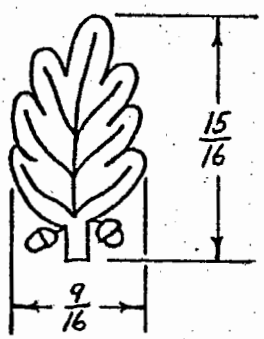
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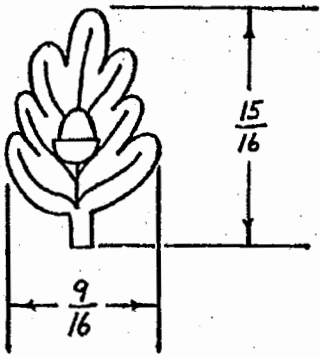
LINE



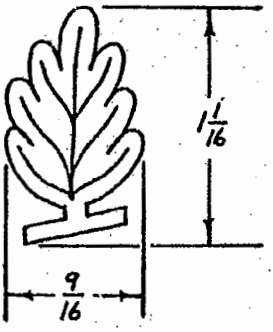
SUPPLY CORPS *



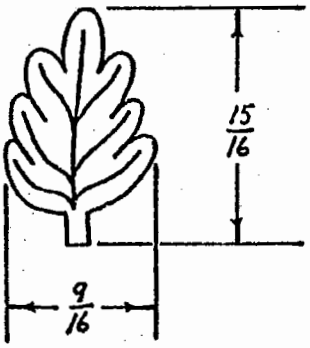
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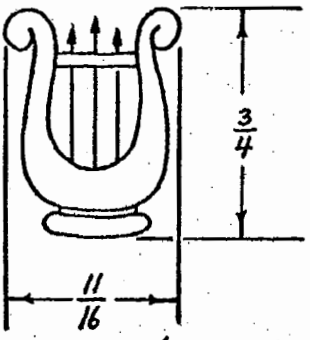
MEDICAL CORPS



MEDICAL SERVICE CORPS *



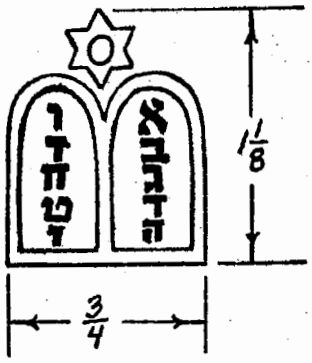
NURSE CORPS



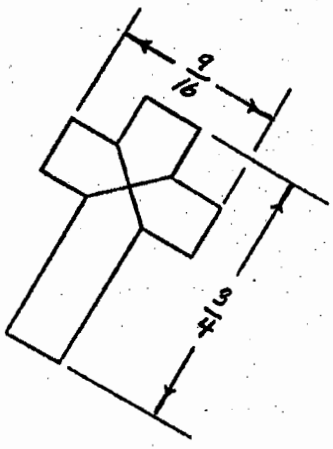
LEADER USN/USNA BAND

FIGURE 5- LINE AND STAFF CORPS DEVICES
CLASS 2 * RIGHT AND LEFT

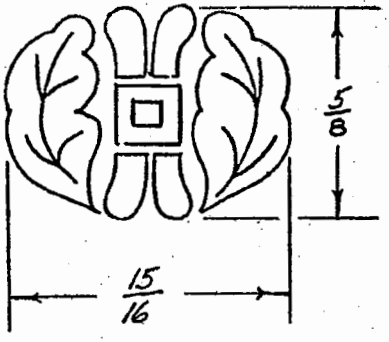
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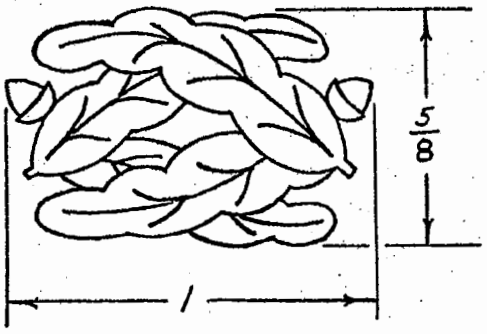
CHAPLAIN CORPS
(JEWISH)



CHAPLAIN CORPS *
(CHRISTIAN)



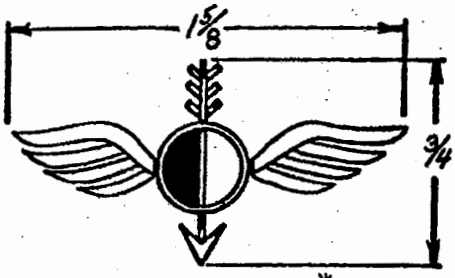
JUDGE ADVOCATE
GENERAL'S CORPS



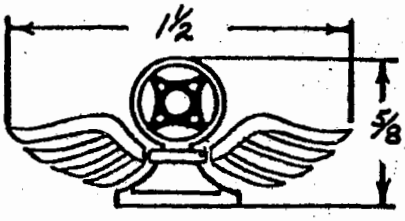
CIVIL ENGINEER CORPS

FIGURE 6 - LINE AND STAFF CORPS DEVICES
CLASS 2 * RIGHT AND LEFT

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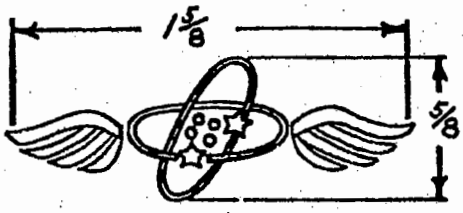
AEROGRAPHER *



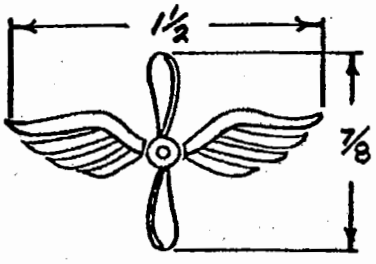
AIR TRAFFIC CONTROL TECHNICIAN



AVIATION BOATSWAIN



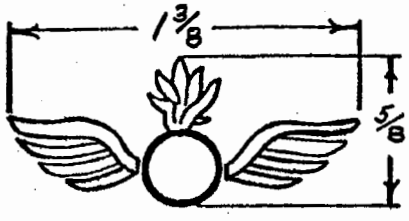
AVIATION ELECTRONICS TECHNICIAN *



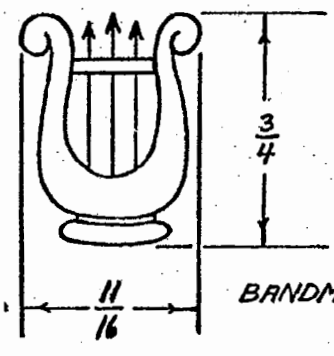
AVIATION MAINTENANCE TECHNICIAN



AVIATION OPERATIONS TECHNICIAN



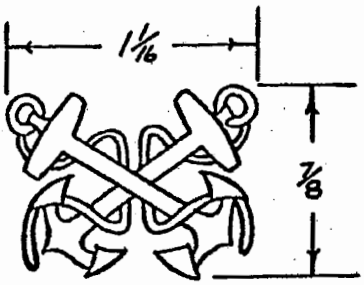
AVIATION ORDNANCE TECHNICIAN
* RIGHT AND LEFT



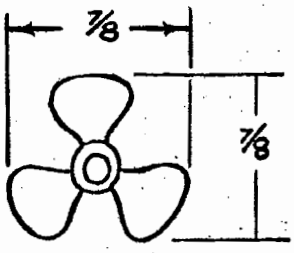
BANDMASTER

FIG. 7 - CLASS 3 LINE AND STAFF CORPS DEVICES

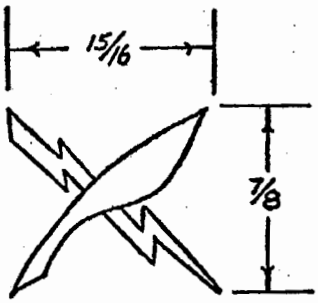
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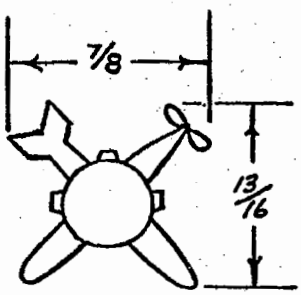
BOATSWAIN



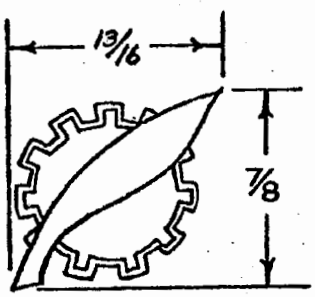
ENGINEERING TECHNICIAN
NUCLEAR POWER TECHNICIAN



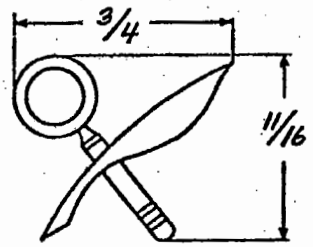
CRYPTOLOGIC TECHNICIAN *



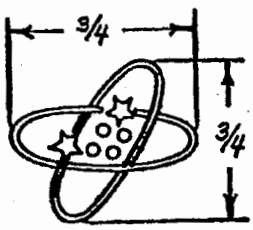
EXPLOSIVE ORDNANCE
DISPOSAL TECHNICIAN *



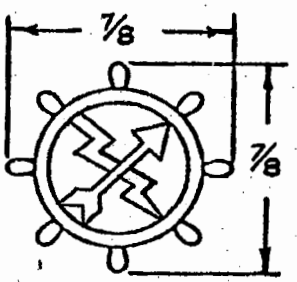
DATA PROCESSING TECHNICIAN *



INTELLIGENCE TECHNICIAN *



ELECTRONICS TECHNICIAN *

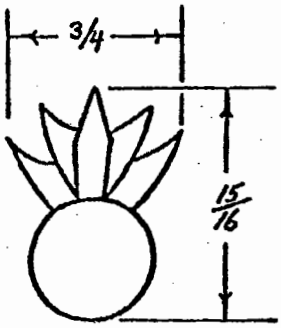


OPERATIONS TECHNICIAN *

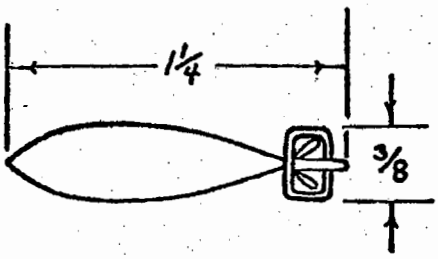
* RIGHT AND LEFT

FIG.8 - CLASS 3 LINE AND STAFF CORPS DEVICES

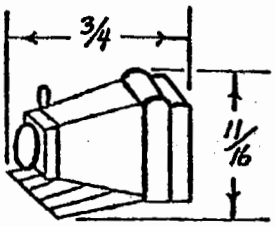
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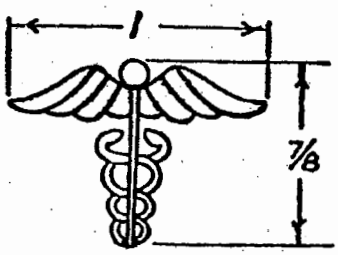
ORDNANCE TECHNICIAN



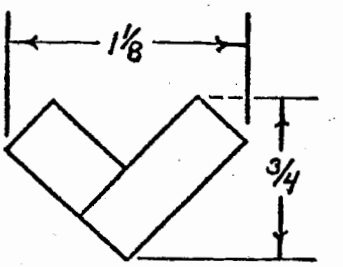
UNDERWATER ORDNANCE TECHNICIAN *



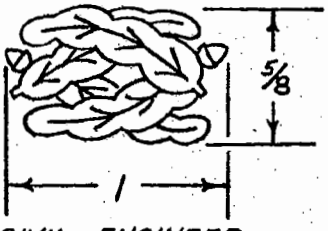
PHOTOGRAPHER *



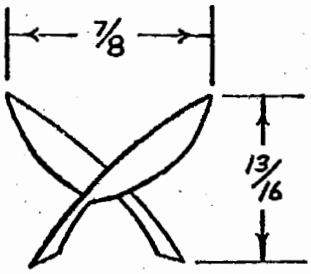
MEDICAL SERVICE
DENTAL SERVICE
PHYSICIAN'S ASSISTANT



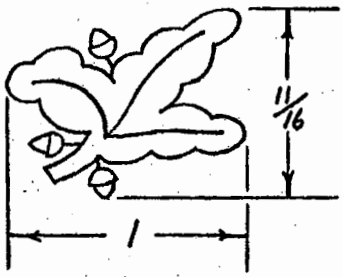
REPAIR TECHNICIAN *



CIVIL ENGINEER



SHIP'S CLERK

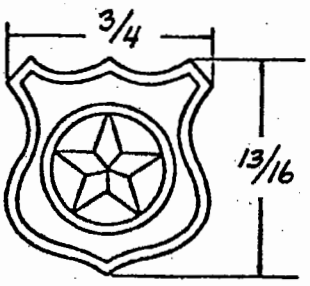


SUPPLY AND FOOD SERVICE *

* RIGHT AND LEFT

FIG. 9 - CLASS 3 LINE AND STAFF CORPS DEVICES

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PHYSICAL SECURITY TECHNICIAN

FIG. 10 - CLASS 3 LINE AND STAFF CORPS DEVICES

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